Medical Knowledge Test Corpus

30-page reference document for evaluating a Speculative RAG system

Each of pages 2-30 covers one clinical topic with a fixed structure: **Overview**, **Diagnosis and Evaluation**, **Management**, and **Key Points**. Every page carries a document ID (MED-02 to MED-30) so retrieval and citation accuracy can be checked against a specific page. Facts are specific (thresholds, drug names, scores, durations), making the corpus suitable for factual question answering, multi-hop questions across pages, and hallucination checks.

Disclaimer: Content is a condensed educational summary based on widely published guidelines and is intended only as test data. It is not a substitute for current clinical guidelines or professional judgment.

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Hypertension

MED-02 | Category: Cardiovascular

Overview

Hypertension is persistently elevated arterial blood pressure and is the leading modifiable risk factor for stroke, myocardial infarction, heart failure, and chronic kidney disease. Under the 2017 ACC/AHA guideline, normal blood pressure is below 120/80 mmHg, elevated is 120-129 systolic with diastolic below 80, stage 1 hypertension is 130-139 or 80-89 mmHg, and stage 2 is 140/90 mmHg or higher. Around 90-95% of cases are primary (essential) hypertension; the remainder are secondary, caused by conditions such as primary aldosteronism, renal artery stenosis, pheochromocytoma, Cushing syndrome, or obstructive sleep apnea.

Diagnosis and Evaluation

Diagnosis requires an average of two or more properly measured readings on at least two occasions. Home blood pressure monitoring and 24-hour ambulatory monitoring help exclude white-coat and masked hypertension. Baseline evaluation includes serum creatinine, electrolytes, fasting glucose, lipid profile, urinalysis, and an electrocardiogram to look for target-organ damage such as left ventricular hypertrophy or proteinuria.

Management

Lifestyle measures include weight loss, the DASH diet, sodium restriction to below 2,300 mg per day, regular aerobic exercise, and limiting alcohol. First-line drug classes are thiazide-type diuretics, ACE inhibitors, angiotensin receptor blockers (ARBs), and calcium channel blockers. ACE inhibitors and ARBs are preferred in patients with diabetes and albuminuria. ACE inhibitors and ARBs must not be combined, and both are contraindicated in pregnancy. Most patients with stage 2 hypertension need two agents from different classes.

Key Points

- Target for most adults: below 130/80 mmHg.
- ACE inhibitors commonly cause a dry cough due to bradykinin accumulation; ARBs do not.
- Hypertensive emergency means severe elevation (often above 180/120 mmHg) with acute organ damage.
- Resistant hypertension is uncontrolled BP despite three drugs including a diuretic.

Type 2 Diabetes Mellitus

MED-03 | Category: Endocrine

Overview

Type 2 diabetes mellitus (T2DM) results from insulin resistance combined with progressive pancreatic beta-cell dysfunction. It accounts for roughly 90-95% of all diabetes. Major risk factors are obesity, physical inactivity, family history, age over 45, prior gestational diabetes, polycystic ovary syndrome, and membership of high-risk ethnic groups. Chronic hyperglycemia causes microvascular complications (retinopathy, nephropathy, neuropathy) and macrovascular complications (coronary disease, stroke, peripheral artery disease).

Diagnosis and Evaluation

Diagnostic criteria from the American Diabetes Association: fasting plasma glucose of 126 mg/dL or higher, 2-hour plasma glucose of 200 mg/dL or higher during a 75 g oral glucose tolerance test, HbA1c of 6.5% or higher, or a random glucose of 200 mg/dL or higher with classic symptoms. Prediabetes is defined as HbA1c 5.7-6.4% or fasting glucose 100-125 mg/dL. Classic symptoms include polyuria, polydipsia, polyphagia, and unexplained weight loss.

Management

Metformin remains the usual first-line drug unless contraindicated (for example eGFR below 30 mL/min/1.73 m2). Metformin works mainly by reducing hepatic glucose production and is weight-neutral with a low hypoglycemia risk. In patients with established atherosclerotic cardiovascular disease, heart failure, or chronic kidney disease, SGLT2 inhibitors (such as empagliflozin) and GLP-1 receptor agonists (such as semaglutide) are recommended for organ protection independent of HbA1c. Sulfonylureas carry a risk of hypoglycemia and weight gain. Insulin is added when glycemic targets are not met.

Key Points

- Typical HbA1c target for most adults: below 7%.
- SGLT2 inhibitors increase the risk of genital mycotic infections and, rarely, euglycemic ketoacidosis.
- Annual screening: dilated eye exam, urine albumin-to-creatinine ratio, foot examination.
- Metformin should be held around iodinated contrast in patients with reduced kidney function.

Type 1 Diabetes and Diabetic Ketoacidosis

MED-04 | Category: Endocrine

Overview

Type 1 diabetes is an autoimmune disease in which T-cell mediated destruction of pancreatic beta cells leads to absolute insulin deficiency. It often presents in childhood or adolescence but can occur at any age. Associated autoantibodies include anti-GAD65, anti-islet cell (ICA), anti-insulin (IAA), and anti-IA-2. It is associated with HLA-DR3 and HLA-DR4 and with other autoimmune diseases such as autoimmune thyroiditis and celiac disease.

Diagnosis and Evaluation

Diabetic ketoacidosis (DKA) is the most serious acute complication. It is characterized by hyperglycemia (typically above 250 mg/dL), an anion-gap metabolic acidosis with arterial pH below 7.30 and bicarbonate below 18 mEq/L, and elevated serum or urine ketones (beta-hydroxybutyrate). Common precipitants are infection, missed insulin doses, and new-onset disease. Symptoms include nausea, vomiting, abdominal pain, Kussmaul respirations, a fruity breath odor, and altered mental status.

Management

DKA management has four pillars: aggressive intravenous fluids (isotonic saline initially), intravenous regular insulin infusion, potassium replacement, and treatment of the precipitating cause. Total-body potassium is depleted even when serum potassium appears normal or high, so insulin must not be started if potassium is below 3.3 mEq/L until it has been repleted. Glucose is monitored hourly; when glucose falls to about 200 mg/dL dextrose is added to fluids to allow insulin to continue clearing ketones. Cerebral edema is a rare but dangerous complication in children.

Key Points

- Basal-bolus insulin or an insulin pump is the standard long-term regimen.
- Continuous glucose monitors improve time-in-range and reduce hypoglycemia.
- Hyperosmolar hyperglycemic state differs from DKA: severe hyperglycemia (above 600 mg/dL), minimal ketosis.
- Hypoglycemia is glucose below 70 mg/dL; treat with 15 g fast-acting carbohydrate (rule of 15).

Coronary Artery Disease and Myocardial Infarction

MED-05 | Category: Cardiovascular

Overview

Coronary artery disease (CAD) is caused mainly by atherosclerotic plaque in the epicardial coronary arteries. Stable plaque produces exertional angina; rupture or erosion of a plaque with thrombus formation produces acute coronary syndrome (ACS), which includes unstable angina, non-ST-elevation myocardial infarction (NSTEMI), and ST-elevation myocardial infarction (STEMI). Risk factors include smoking, hypertension, dyslipidemia, diabetes, obesity, family history of premature CAD, and chronic kidney disease.

Diagnosis and Evaluation

Typical ACS presentation is crushing substernal chest pressure radiating to the left arm or jaw with diaphoresis and dyspnea; women, older adults, and patients with diabetes may present atypically. A 12-lead ECG should be obtained within 10 minutes of arrival. STEMI is diagnosed by new ST elevation at the J point in two contiguous leads (at least 1 mm in most leads) or a new left bundle branch block with a compatible history. Cardiac troponin I or T is the preferred biomarker; it rises 3-4 hours after injury and may remain elevated for 7-14 days.

Management

Initial treatment of ACS includes aspirin 162-325 mg chewed, a P2Y12 inhibitor (clopidogrel, ticagrelor, or prasugrel), anticoagulation with heparin, a high-intensity statin, and a beta-blocker if no contraindication. Nitroglycerin relieves ischemic pain but is contraindicated after recent phosphodiesterase-5 inhibitor use and in right ventricular infarction. For STEMI, primary percutaneous coronary intervention (PCI) is preferred with a goal door-to-balloon time of 90 minutes or less; if PCI is unavailable, fibrinolysis should begin within 30 minutes of arrival.

Key Points

- Inferior STEMI (leads II, III, aVF) usually involves the right coronary artery.
- Dual antiplatelet therapy is generally continued for 12 months after ACS.
- Cardiac rehabilitation lowers mortality and readmission after MI.
- Complications include arrhythmias, cardiogenic shock, papillary muscle rupture, and ventricular free-wall rupture.

Heart Failure

MED-06 | Category: Cardiovascular

Overview

Heart failure (HF) is a clinical syndrome in which the heart cannot pump enough blood to meet metabolic demands or can do so only at elevated filling pressures. It is classified by left ventricular ejection fraction (LVEF): HF with reduced EF (HFrEF) is LVEF of 40% or less, HF with mildly reduced EF is 41-49%, and HF with preserved EF (HFpEF) is 50% or more. Common causes are ischemic heart disease, hypertension, cardiomyopathy, valvular disease, and chronic alcohol use.

Diagnosis and Evaluation

Symptoms include exertional dyspnea, orthopnea, paroxysmal nocturnal dyspnea, fatigue, and peripheral edema. Examination may show elevated jugular venous pressure, an S3 gallop, pulmonary crackles, and hepatomegaly. B-type natriuretic peptide (BNP) and NT-proBNP support the diagnosis, and transthoracic echocardiography establishes the ejection fraction. The ACC/AHA staging system runs from Stage A (at risk) to Stage D (refractory HF); the NYHA functional class runs from I (no limitation) to IV (symptoms at rest).

Management

Guideline-directed medical therapy for HFrEF consists of four drug classes: an ARNI (sacubitril/valsartan) or ACE inhibitor/ARB, an evidence-based beta-blocker (carvedilol, metoprolol succinate, or bisoprolol), a mineralocorticoid receptor antagonist (spironolactone or eplerenone), and an SGLT2 inhibitor (dapagliflozin or empagliflozin). Loop diuretics relieve congestion but do not improve survival. Devices include an implantable cardioverter-defibrillator for LVEF of 35% or less despite therapy, and cardiac resynchronization therapy for wide QRS with left bundle branch block.

Key Points

- Sacubitril/valsartan must not be given within 36 hours of an ACE inhibitor (angioedema risk).
- Monitor potassium and creatinine with MRAs and ACE inhibitors/ARNIs.
- Non-dihydropyridine calcium channel blockers are avoided in HFrEF.
- Daily weights; a gain of more than 2-3 pounds in a day or 5 pounds in a week should prompt a call to the care team.

Atrial Fibrillation

MED-07 | Category: Cardiovascular

Overview

Atrial fibrillation (AF) is the most common sustained cardiac arrhythmia, characterized by chaotic atrial electrical activity and an irregularly irregular ventricular response. It is classified as paroxysmal (terminates within 7 days), persistent (more than 7 days), long-standing persistent (more than 12 months), or permanent. Risk factors include advancing age, hypertension, heart failure, valvular disease (especially mitral stenosis), hyperthyroidism, obesity, sleep apnea, and alcohol ("holiday heart").

Diagnosis and Evaluation

The ECG shows absent P waves, an irregularly irregular rhythm, and fibrillatory baseline waves. Patients may report palpitations, dyspnea, fatigue, or presyncope, or may be asymptomatic. Initial work-up includes thyroid function tests, electrolytes, and an echocardiogram. Stroke risk is estimated with the CHA2DS2-VASc score, which assigns points for congestive heart failure, hypertension, age 75 or older (2 points), diabetes, prior stroke or TIA (2 points), vascular disease, age 65-74, and female sex.

Management

Management addresses three goals: stroke prevention, rate control, and rhythm control. Direct oral anticoagulants (apixaban, rivaroxaban, edoxaban, dabigatran) are preferred over warfarin in non-valvular AF; warfarin remains required for moderate-to-severe mitral stenosis and mechanical heart valves. Rate control uses beta-blockers or non-dihydropyridine calcium channel blockers (diltiazem, verapamil) with a lenient target resting heart rate below 110 bpm. Rhythm control options include electrical cardioversion, antiarrhythmics (amiodarone, flecainide, sotalol), and catheter ablation with pulmonary vein isolation. Hemodynamically unstable AF requires immediate synchronized cardioversion.

Key Points

- Anticoagulate if CHA2DS2-VASc is 2 or more in men, or 3 or more in women.
- HAS-BLED estimates bleeding risk on anticoagulation.
- Cardioversion in AF lasting more than 48 hours requires 3 weeks of anticoagulation first or a transesophageal echo to exclude atrial thrombus.
- Flecainide is avoided in structural heart disease or coronary disease.

Ischemic Stroke

MED-08 | Category: Neurology

Overview

Stroke is a sudden focal neurological deficit caused by disrupted cerebral blood flow. About 87% of strokes are ischemic and the rest hemorrhagic (intracerebral or subarachnoid). Ischemic stroke is caused by large-artery atherosclerosis, cardioembolism (often from atrial fibrillation), small-vessel (lacunar) disease, or cryptogenic mechanisms. A transient ischemic attack (TIA) produces symptoms that resolve, usually within an hour, without infarction on imaging.

Diagnosis and Evaluation

Recognition uses the FAST mnemonic: Face drooping, Arm weakness, Speech difficulty, Time to call emergency services. An urgent non-contrast head CT excludes hemorrhage; CT angiography identifies large-vessel occlusion, and MRI with diffusion-weighted imaging is the most sensitive method for early infarct. The NIH Stroke Scale (NIHSS, 0-42) quantifies severity. Fingerstick glucose must be checked because hypoglycemia can mimic stroke.

Management

Intravenous thrombolysis with alteplase or tenecteplase is indicated within 4.5 hours of last known well in eligible patients; contraindications include intracranial hemorrhage, recent major surgery, and uncontrolled severe hypertension (above 185/110 mmHg before treatment). Mechanical thrombectomy is recommended for large-vessel occlusion of the internal carotid or proximal middle cerebral artery, up to 24 hours in selected patients based on imaging. Secondary prevention includes antiplatelet therapy, high-intensity statin, blood pressure control, anticoagulation for AF, and carotid endarterectomy for symptomatic carotid stenosis of 70-99%.

Key Points

- Time is brain: about 1.9 million neurons are lost each minute in untreated large-vessel stroke.
- Aspirin is withheld for 24 hours after thrombolysis.
- Dysphagia screening must precede oral intake.
- Permissive hypertension up to 220/120 mmHg is allowed if no thrombolysis is given.

Chronic Obstructive Pulmonary Disease

MED-09 | Category: Pulmonology

Overview

Chronic obstructive pulmonary disease (COPD) is a progressive disease defined by persistent airflow limitation due to airway and alveolar abnormalities, usually caused by cigarette smoking. It encompasses chronic bronchitis (productive cough for at least 3 months in 2 consecutive years) and emphysema (destruction of alveolar walls). Alpha-1 antitrypsin deficiency should be suspected in young patients, nonsmokers, or those with basal-predominant emphysema.

Diagnosis and Evaluation

Spirometry confirms the diagnosis: a post-bronchodilator FEV1/FVC ratio below 0.70. The GOLD system grades severity by FEV1 percent predicted: GOLD 1 (mild) 80% or more, GOLD 2 (moderate) 50-79%, GOLD 3 (severe) 30-49%, and GOLD 4 (very severe) below 30%. Patients are further grouped (A, B, E) by symptom burden and exacerbation history. Chest imaging may show hyperinflation and flattened diaphragms.

Management

Smoking cessation is the single most effective intervention to slow decline. Long-acting bronchodilators (LABA and LAMA) are the cornerstone of maintenance therapy; inhaled corticosteroids are added for frequent exacerbations and blood eosinophils of 300 cells/microL or more. Short-acting beta-agonists are used for rescue. Long-term oxygen therapy improves survival when resting PaO2 is 55 mmHg or lower (or SpO2 88% or lower). Acute exacerbations are treated with bronchodilators, a 5-day course of oral prednisone 40 mg, and antibiotics when sputum purulence and increased volume or dyspnea are present. Non-invasive ventilation reduces intubation and mortality in hypercapnic respiratory failure.

Key Points

- Target oxygen saturation in COPD exacerbation: 88-92%.
- Vaccination against influenza, pneumococcus, COVID-19, and RSV is recommended.
- Pulmonary rehabilitation improves exercise capacity and quality of life.
- Roflumilast may reduce exacerbations in chronic bronchitis with FEV1 below 50%.

Asthma

MED-10 | Category: Pulmonology

Overview

Asthma is a chronic inflammatory disorder of the airways causing variable airflow obstruction, bronchial hyperresponsiveness, and recurrent wheeze, cough, chest tightness, and dyspnea, often worse at night or early morning. Triggers include allergens, viral infections, exercise, cold air, smoke, and aspirin or NSAIDs in susceptible patients. It is frequently associated with atopy, allergic rhinitis, and eczema, and with T-helper 2 inflammation involving IL-4, IL-5, and IL-13.

Diagnosis and Evaluation

Diagnosis rests on history plus demonstration of reversible obstruction: an increase in FEV1 of at least 12% and 200 mL after a bronchodilator, or positive bronchial challenge (methacholine). Peak expiratory flow monitoring shows diurnal variability. Fractional exhaled nitric oxide (FeNO) and blood eosinophils help identify type 2 inflammation.

Management

The current GINA strategy recommends as-needed low-dose inhaled corticosteroid-formoterol as the preferred reliever across steps, rather than a short-acting beta-agonist alone, because SABA-only treatment increases exacerbation risk. Maintenance therapy steps up from low-dose ICS to ICS-LABA, medium or high-dose ICS-LABA, and then add-on LAMA. For severe eosinophilic asthma, biologics include omalizumab (anti-IgE), mepolizumab and benralizumab (anti-IL-5 pathway), and dupilumab (anti-IL-4 receptor alpha). Acute severe asthma requires oxygen, repeated inhaled albuterol with ipratropium, systemic corticosteroids, and consideration of intravenous magnesium sulfate.

Key Points

- A silent chest during an attack is an ominous sign of near-fatal obstruction.
- Beta-blockers can precipitate bronchospasm and are generally avoided.
- Inhaler technique and a written asthma action plan reduce exacerbations.
- Leukotriene receptor antagonists (montelukast) carry a boxed warning for neuropsychiatric events.

Community-Acquired Pneumonia

MED-11 | Category: Infectious Disease

Overview

Community-acquired pneumonia (CAP) is infection of the lung parenchyma acquired outside of hospital settings. Streptococcus pneumoniae is the most common bacterial cause. Other causes include Haemophilus influenzae, Mycoplasma pneumoniae, Chlamydophila pneumoniae, Legionella pneumophila, Staphylococcus aureus, and respiratory viruses such as influenza, SARS-CoV-2, and RSV. Aspiration pneumonia occurs in patients with impaired swallowing or consciousness.

Diagnosis and Evaluation

Patients present with fever, productive cough, pleuritic chest pain, and dyspnea; examination may reveal crackles, bronchial breath sounds, and dullness to percussion. A chest radiograph showing a new infiltrate confirms the diagnosis. Severity is assessed with CURB-65 (Confusion, Urea above 7 mmol/L, Respiratory rate 30 or more, low Blood pressure, age 65 or older) or the Pneumonia Severity Index. A CURB-65 score of 0-1 supports outpatient care; 2 suggests hospital admission; 3 or more suggests ICU consideration.

Management

For healthy outpatients without comorbidities, guidelines recommend amoxicillin, doxycycline, or a macrolide where local pneumococcal macrolide resistance is below 25%. For outpatients with comorbidities, use combination therapy with amoxicillin-clavulanate or a cephalosporin plus a macrolide or doxycycline, or monotherapy with a respiratory fluoroquinolone (levofloxacin or moxifloxacin). Hospitalized non-ICU patients receive a beta-lactam plus a macrolide, or a respiratory fluoroquinolone alone. Typical treatment duration is a minimum of 5 days, continuing until clinically stable. Pneumococcal conjugate vaccination reduces invasive disease in older adults and children.

Key Points

- Legionella is diagnosed with urinary antigen testing.
- Complications include parapneumonic effusion, empyema, lung abscess, and sepsis.
- Pneumonia that fails to improve in 48-72 hours should prompt re-evaluation for resistance or complications.
- Rusty-colored sputum is classically associated with pneumococcal pneumonia.

Tuberculosis

MED-12 | Category: Infectious Disease

Overview

Tuberculosis (TB) is caused by Mycobacterium tuberculosis, an acid-fast bacillus transmitted by airborne droplet nuclei. It remains one of the leading infectious causes of death worldwide. Most infections are latent (no symptoms, not contagious); about 5-10% of immunocompetent people with latent TB develop active disease over a lifetime, with a much higher risk in people with HIV, diabetes, malnutrition, or those on TNF-alpha inhibitors.

Diagnosis and Evaluation

Active pulmonary TB causes chronic cough, hemoptysis, fever, night sweats, and weight loss. Chest radiography typically shows upper-lobe cavitary lesions in reactivation disease. Diagnosis requires sputum smear microscopy for acid-fast bacilli, culture on Lowenstein-Jensen medium or liquid media, and nucleic-acid amplification such as Xpert MTB/RIF, which also detects rifampin resistance. Latent TB infection is detected with the tuberculin skin test or an interferon-gamma release assay (IGRA); neither distinguishes latent from active disease.

Management

Standard treatment for drug-susceptible TB is a 6-month regimen: an intensive 2-month phase of isoniazid, rifampin, pyrazinamide, and ethambutol (RIPE), followed by a 4-month continuation phase of isoniazid and rifampin. Pyridoxine (vitamin B6) is given with isoniazid to prevent peripheral neuropathy. Known adverse effects: isoniazid causes hepatotoxicity; rifampin causes orange discoloration of body fluids and induces cytochrome P450; pyrazinamide causes hyperuricemia; ethambutol causes optic neuritis. Multidrug-resistant TB (resistance to at least isoniazid and rifampin) is treated with regimens containing bedaquiline, pretomanid, and linezolid.

Key Points

- Directly observed therapy improves adherence.
- Patients with active pulmonary TB require airborne isolation.
- Test for HIV in every patient diagnosed with TB.
- BCG vaccine can cause false-positive tuberculin skin tests; IGRAs are not affected.

Chronic Kidney Disease

MED-13 | Category: Nephrology

Overview

Chronic kidney disease (CKD) is defined as abnormalities of kidney structure or function present for more than 3 months with implications for health. The leading causes are diabetes mellitus and hypertension, followed by glomerulonephritis, polycystic kidney disease, and chronic tubulointerstitial disease. CKD is staged by estimated glomerular filtration rate (eGFR) and by albuminuria category.

Diagnosis and Evaluation

The stages by eGFR (mL/min/1.73 m2) are: G1 90 or higher with kidney damage, G2 60-89, G3a 45-59, G3b 30-44, G4 15-29, and G5 below 15 (kidney failure). Albuminuria is categorized by urine albumin-to-creatinine ratio: A1 below 30 mg/g, A2 30-300 mg/g, and A3 above 300 mg/g. Complications include anemia (reduced erythropoietin), secondary hyperparathyroidism with hyperphosphatemia and hypocalcemia, metabolic acidosis, hyperkalemia, volume overload, and increased cardiovascular risk.

Management

Treatment aims to slow progression: strict blood pressure control (generally below 120 mmHg systolic if tolerated per KDIGO), an ACE inhibitor or ARB for albuminuria, an SGLT2 inhibitor (shown to slow CKD progression with or without diabetes), and finerenone for diabetic CKD with albuminuria. Nephrotoxins such as NSAIDs and unnecessary contrast should be avoided, and drugs dosed for kidney function. Anemia is treated with iron and erythropoiesis-stimulating agents; phosphate binders and vitamin D analogs manage mineral bone disease. Kidney replacement therapy (hemodialysis, peritoneal dialysis, or transplantation) is considered at eGFR below 15 or earlier for uremic symptoms; transplantation provides the best survival and quality of life.

Key Points

- Cystatin C can refine eGFR estimates when creatinine is unreliable.
- Metformin dose is reduced at eGFR 30-45 and stopped below 30.
- Uremic complications include pericarditis, encephalopathy, and bleeding from platelet dysfunction.
- Hyperkalemia with ECG changes needs intravenous calcium gluconate for membrane stabilization.

Acute Kidney Injury

MED-14 | Category: Nephrology

Overview

Acute kidney injury (AKI) is a rapid decline in kidney function. KDIGO criteria define AKI as a rise in serum creatinine of 0.3 mg/dL or more within 48 hours, an increase to 1.5 times baseline within 7 days, or urine output below 0.5 mL/kg/hour for 6 hours. Causes are classified as prerenal (reduced perfusion, such as hypovolemia, heart failure, or sepsis), intrinsic renal (acute tubular necrosis, acute interstitial nephritis, glomerulonephritis), and postrenal (urinary tract obstruction).

Diagnosis and Evaluation

Evaluation includes history of volume loss and nephrotoxic drugs, urinalysis with microscopy, urine electrolytes, and renal ultrasound to exclude obstruction. Prerenal azotemia typically has a BUN-to-creatinine ratio above 20, a fractional excretion of sodium (FeNa) below 1%, and bland sediment. Acute tubular necrosis shows muddy brown granular casts and FeNa above 2%. Acute interstitial nephritis, often drug-induced (beta-lactams, proton pump inhibitors, NSAIDs), may show white-cell casts and eosinophiluria. Red-cell casts indicate glomerulonephritis.

Management

Management is directed at the cause: restore volume with balanced crystalloids in hypovolemia, relieve obstruction with catheter or nephrostomy, stop nephrotoxins, and adjust drug doses. Indications for urgent dialysis follow the mnemonic AEIOU: Acidosis refractory to treatment, Electrolyte abnormalities (especially refractory hyperkalemia), Intoxications (certain toxic alcohols, lithium, salicylates), Overload of fluid unresponsive to diuretics, and Uremic complications such as pericarditis or encephalopathy. Contrast-associated AKI is reduced by hydration and minimizing contrast volume.

Key Points

- Rhabdomyolysis causes AKI with markedly elevated creatine kinase and dark urine.
- Diuretics do not improve AKI outcomes and are used only for volume overload.
- AKI increases the risk of subsequent CKD.
- Hepatorenal syndrome is functional AKI in advanced cirrhosis, treated with albumin and vasoconstrictors such as terlipressin.

Cirrhosis and Portal Hypertension

MED-15 | Category: Hepatology

Overview

Cirrhosis is the end stage of chronic liver injury, characterized by diffuse fibrosis and regenerative nodules that distort hepatic architecture. Leading causes are chronic hepatitis C, alcohol-associated liver disease, metabolic dysfunction-associated steatotic liver disease (MASLD), and chronic hepatitis B; less common causes include autoimmune hepatitis, primary biliary cholangitis, primary sclerosing cholangitis, hemochromatosis, and Wilson disease. Compensated cirrhosis may be asymptomatic, while decompensation is marked by ascites, variceal bleeding, hepatic encephalopathy, or jaundice.

Diagnosis and Evaluation

Physical findings include spider angiomata, palmar erythema, gynecomastia, splenomegaly, caput medusae, and asterixis. Laboratory abnormalities include thrombocytopenia (hypersplenism), prolonged INR, low albumin, and elevated bilirubin. Severity is graded by the Child-Pugh score (classes A, B, C) and the MELD score, which uses bilirubin, INR, creatinine, and sodium and guides liver transplant allocation. Noninvasive fibrosis assessment includes transient elastography and the FIB-4 index.

Management

Ascites is treated with sodium restriction and spironolactone plus furosemide; large-volume paracentesis is accompanied by albumin infusion. A diagnostic paracentesis with ascitic neutrophil count of 250 cells/mm3 or higher diagnoses spontaneous bacterial peritonitis (SBP), treated with cefotaxime or ceftriaxone plus albumin; prophylaxis follows an episode. Variceal bleeding is managed with resuscitation, intravenous octreotide, prophylactic antibiotics, and urgent endoscopic band ligation; nonselective beta-blockers (propranolol, nadolol, carvedilol) prevent first bleeding. Hepatic encephalopathy is treated with lactulose and rifaximin. Hepatocellular carcinoma surveillance with ultrasound every 6 months is recommended. Liver transplantation is the definitive treatment for decompensated disease.

Key Points

- Serum-ascites albumin gradient of 1.1 g/dL or higher indicates portal hypertension.
- Avoid NSAIDs and sedatives in cirrhosis.
- Alcohol cessation can stabilize or improve alcohol-related cirrhosis.
- Vaccination against hepatitis A and B is recommended in chronic liver disease.

Viral Hepatitis B and C

MED-16 | Category: Hepatology

Overview

Hepatitis B virus (HBV) is a DNA virus transmitted by blood, sexual contact, and perinatally; it is a major cause of cirrhosis and hepatocellular carcinoma worldwide. Hepatitis C virus (HCV) is an RNA virus transmitted mainly through blood, most often injection drug use. Progression to chronic infection occurs in about 90% of infants infected perinatally with HBV, but only around 5% of adults, whereas about 70-85% of HCV infections become chronic.

Diagnosis and Evaluation

HBV serology: HBsAg indicates current infection; anti-HBs indicates immunity from vaccination or recovery; anti-HBc IgM suggests acute infection; total anti-HBc with negative HBsAg and positive anti-HBs indicates resolved infection; HBeAg and high HBV DNA indicate active replication and infectivity. HBsAg persisting beyond 6 months defines chronic hepatitis B. HCV screening uses anti-HCV antibody, confirmed with HCV RNA PCR; genotyping is less important now with pangenotypic regimens.

Management

Chronic HBV is treated with nucleos(t)ide analogues such as entecavir or tenofovir, which suppress viral replication but rarely cure; pegylated interferon is an alternative in selected patients. Patients receiving immunosuppressive or chemotherapy regimens need HBV screening and prophylaxis because of reactivation risk. HCV is now curable: direct-acting antivirals such as sofosbuvir/velpatasvir or glecaprevir/pibrentasvir given for 8-12 weeks achieve sustained virologic response (SVR12, undetectable RNA 12 weeks after therapy) in over 95% of patients. HBV vaccination (three-dose series, or two-dose Heplisav-B) is highly effective; no vaccine exists for HCV.

Key Points

- Universal HBV vaccination is recommended for infants and for adults 19-59 years.
- HBV and HCV are screened in pregnancy; infants of HBsAg-positive mothers receive HBIG plus vaccine within 12 hours of birth.
- Hepatitis A and E are fecal-oral and do not cause chronic infection (except HEV in immunosuppressed).
- HBV/HDV coinfection worsens liver disease.

GERD and Peptic Ulcer Disease

MED-17 | Category: Gastroenterology

Overview

Gastroesophageal reflux disease (GERD) results from reflux of gastric contents causing troublesome symptoms or complications, mainly through transient lower esophageal sphincter relaxation. Typical symptoms are heartburn and regurgitation; atypical presentations include chronic cough, hoarseness, and non-cardiac chest pain. Complications include erosive esophagitis, strictures, Barrett esophagus (intestinal metaplasia with risk of adenocarcinoma), and adenocarcinoma. Peptic ulcer disease (PUD) is a break in the gastric or duodenal mucosa, mostly caused by Helicobacter pylori infection or NSAID use.

Diagnosis and Evaluation

Alarm features that warrant endoscopy include dysphagia, odynophagia, weight loss, gastrointestinal bleeding, anemia, persistent vomiting, and age older than 60 with new symptoms. A trial of proton pump inhibitor (PPI) therapy is reasonable for uncomplicated typical GERD. H. pylori testing uses the urea breath test, stool antigen test, or biopsy urease test; PPIs must be held for 2 weeks and antibiotics for 4 weeks before testing to avoid false negatives. Duodenal ulcer pain often improves with meals and occurs at night, while gastric ulcer pain may worsen with eating; gastric ulcers require biopsy to exclude malignancy.

Management

PPIs (omeprazole, esomeprazole, pantoprazole) are the most effective acid suppressants and are given 30-60 minutes before a meal for 8 weeks in erosive esophagitis. Lifestyle measures include weight loss, head-of-bed elevation, and avoiding late meals. Anti-reflux surgery (fundoplication) is an option for refractory disease. H. pylori eradication uses bismuth quadruple therapy (PPI, bismuth, tetracycline, metronidazole) for 14 days, or in some regions a clarithromycin-based triple therapy where resistance is low. NSAID-related ulcers are treated by stopping the NSAID and giving a PPI. Upper GI bleeding from an ulcer is treated with intravenous PPI and endoscopic hemostasis.

Key Points

- Zollinger-Ellison syndrome (gastrinoma) causes refractory, multiple ulcers with high serum gastrin.
- Long-term PPI use is associated with B12 deficiency, hypomagnesemia, fractures, and C. difficile infection.
- Perforated ulcer causes sudden severe abdominal pain with free air under the diaphragm.
- Barrett esophagus requires surveillance endoscopy.

Inflammatory Bowel Disease

MED-18 | Category: Gastroenterology

Overview

Inflammatory bowel disease (IBD) comprises Crohn disease and ulcerative colitis (UC), chronic immune-mediated conditions of the gastrointestinal tract. Crohn disease can affect any part of the gut from mouth to anus, most commonly the terminal ileum, with skip lesions, transmural inflammation, and non-caseating granulomas. UC is limited to the colon and rectum, begins in the rectum, spreads continuously, and affects only the mucosa and submucosa. Smoking increases risk of Crohn disease but is protective against UC.

Diagnosis and Evaluation

Crohn disease typically causes abdominal pain, diarrhea, weight loss, perianal fistulas, and strictures; UC causes bloody diarrhea with urgency and tenesmus. Extraintestinal manifestations include arthritis, erythema nodosum, pyoderma gangrenosum, uveitis, and primary sclerosing cholangitis (more common in UC). Diagnosis relies on colonoscopy with biopsy, imaging (MR enterography or CT enterography), fecal calprotectin as a marker of intestinal inflammation, and exclusion of infection including Clostridioides difficile.

Management

Mild-to-moderate UC is treated with 5-aminosalicylates (mesalamine), rectal therapy for distal disease, and short courses of corticosteroids for flares. Moderate-to-severe IBD uses biologic therapy: anti-TNF agents (infliximab, adalimumab), the anti-integrin vedolizumab, anti-IL-12/23 ustekinumab, anti-IL-23 agents (risankizumab, mirikizumab), and small molecules such as JAK inhibitors (tofacitinib, upadacitinib) and S1P modulators (ozanimod). Thiopurines (azathioprine) are used for maintenance. Before biologics, screen for latent TB and hepatitis B. Severe acute colitis requires hospitalization with intravenous corticosteroids; colectomy is curative for UC. Surgery in Crohn disease is not curative.

Key Points

- Colorectal cancer surveillance colonoscopy begins about 8 years after UC or Crohn colitis diagnosis.
- Toxic megacolon (colon diameter above 6 cm) is a life-threatening complication.
- Smoking cessation is essential in Crohn disease.
- Corticosteroids are not used for long-term maintenance.

Anemia: Iron Deficiency and Vitamin B12 Deficiency

MED-19 | Category: Hematology

Overview

Anemia is a reduction in hemoglobin below the reference range (generally below 13.0 g/dL in men and 12.0 g/dL in non-pregnant women). Classification by mean corpuscular volume (MCV) guides differential diagnosis: microcytic (MCV below 80 fL: iron deficiency, thalassemia, anemia of chronic disease, sideroblastic anemia), normocytic (80-100 fL: acute blood loss, hemolysis, chronic kidney disease, anemia of chronic disease), and macrocytic (above 100 fL: vitamin B12 or folate deficiency, alcohol, hypothyroidism, myelodysplasia, certain drugs).

Diagnosis and Evaluation

Iron deficiency anemia, the most common anemia worldwide, shows low ferritin (below 30 ng/mL, often below 15), low serum iron, high total iron-binding capacity, low transferrin saturation, and elevated red cell distribution width. In adult men and postmenopausal women, gastrointestinal blood loss must be excluded with endoscopy and colonoscopy. Vitamin B12 deficiency, caused by pernicious anemia (autoimmune loss of intrinsic factor), gastrectomy, ileal disease, metformin, or strict veganism, produces macro-ovalocytes, hypersegmented neutrophils, elevated methylmalonic acid and homocysteine, and neurologic features such as paresthesias, loss of vibration sense, and subacute combined degeneration of the spinal cord.

Management

Iron deficiency is treated with oral ferrous sulfate (often once daily or on alternate days for better absorption and tolerance) with a rise in hemoglobin of about 1 g/dL per week to two weeks; intravenous iron (ferric carboxymaltose, iron sucrose) is used for intolerance, malabsorption, or ongoing losses. Vitamin B12 deficiency is treated with intramuscular cyanocobalamin or high-dose oral therapy; folate must not be given alone in B12 deficiency because it can correct anemia while neurologic damage progresses. Transfusion is reserved for symptomatic or hemodynamically unstable patients, generally at a hemoglobin threshold of 7 g/dL (7-8 g/dL in cardiovascular disease).

Key Points

- Reticulocyte count distinguishes underproduction from hemolysis or bleeding.
- Thalassemia trait has a low MCV with a normal or high red cell count and normal ferritin.
- Anti-intrinsic factor antibodies are specific for pernicious anemia.
- Lead poisoning and sideroblastic anemia show basophilic stippling or ring sideroblasts.

Sepsis and Septic Shock

MED-20 | Category: Critical Care

Overview

Sepsis is defined by the Sepsis-3 consensus as life-threatening organ dysfunction caused by a dysregulated host response to infection, identified by an acute rise of 2 points or more in the Sequential Organ Failure Assessment (SOFA) score. Septic shock is sepsis with persistent hypotension requiring vasopressors to maintain mean arterial pressure (MAP) of 65 mmHg or more and a serum lactate above 2 mmol/L despite adequate fluid resuscitation. Common sources are the lungs, urinary tract, abdomen, and skin or soft tissue, and bloodstream infections.

Diagnosis and Evaluation

The quick SOFA (qSOFA) screening tool awards one point each for respiratory rate of 22 or more, altered mentation, and systolic blood pressure of 100 mmHg or lower; a score of 2 or more suggests higher risk of poor outcome. Blood cultures should be obtained before antibiotics when this does not delay treatment, along with lactate, complete blood count, metabolic panel, coagulation studies, and cultures from suspected sources. Procalcitonin can support antibiotic decisions but should not replace clinical judgment.

Management

The Surviving Sepsis Campaign recommends rapid recognition and treatment: administer broad-spectrum antimicrobials within 1 hour of recognition for septic shock or probable sepsis, give at least 30 mL/kg of intravenous crystalloid within the first 3 hours for sepsis-induced hypoperfusion (balanced crystalloids preferred), and start norepinephrine as the first-line vasopressor, with vasopressin added if needed. Hydrocortisone is considered if vasopressor requirements remain high. Source control (drainage of abscess, removal of infected devices, surgery for perforation) should occur as soon as medically and logistically practical. Antibiotics are narrowed once cultures return.

Key Points

- Each hour of delay in antibiotics in septic shock is associated with increased mortality.
- Lactate clearance and capillary refill time can guide resuscitation.
- Disseminated intravascular coagulation may complicate sepsis.
- Neutropenic fever requires immediate empiric antipseudomonal beta-lactam therapy.

Antimicrobial Resistance and Stewardship

MED-21 | Category: Infectious Disease

Overview

Antimicrobial resistance (AMR) occurs when microorganisms evolve to survive drugs that once killed them, and is driven by overuse and misuse of antibiotics in humans, animals, and agriculture. Mechanisms include beta-lactamase production (including extended-spectrum beta-lactamases, ESBLs, and carbapenemases), altered penicillin-binding proteins (as in methicillin-resistant Staphylococcus aureus, MRSA, via the mecA gene), efflux pumps, target modification, and reduced permeability. WHO priority pathogens include carbapenem-resistant Acinetobacter baumannii, Pseudomonas aeruginosa, and Enterobacterales.

Diagnosis and Evaluation

Important resistant organisms include MRSA, vancomycin-resistant Enterococcus (VRE), ESBL-producing and carbapenem-resistant Enterobacterales (CRE), multidrug-resistant Pseudomonas, and drug-resistant Mycobacterium tuberculosis. Susceptibility is reported by minimum inhibitory concentration (MIC) using breakpoints from CLSI or EUCAST. Clostridioides difficile infection is a major consequence of antibiotic exposure, especially clindamycin, fluoroquinolones, and cephalosporins.

Management

Antibiotic stewardship programs aim to optimize therapy. Core strategies include the 'Start Smart then Focus' approach: obtaining cultures before therapy, using narrow-spectrum agents when possible, de-escalating after culture results, using the shortest effective duration, converting intravenous to oral therapy promptly, and avoiding treatment of asymptomatic bacteriuria. Treatment of MRSA infections uses vancomycin, linezolid, daptomycin (not for pneumonia), or ceftaroline; ESBL infections are treated with carbapenems; CRE may need newer agents such as ceftazidime-avibactam or meropenem-vaborbactam. Infection prevention measures include hand hygiene, contact precautions, environmental cleaning, and vaccination.

Key Points

- Most upper respiratory tract infections are viral and do not need antibiotics.
- Therapeutic drug monitoring is used for vancomycin and aminoglycosides.
- Daptomycin is inactivated by pulmonary surfactant.
- C. difficile first-line therapy is oral fidaxomicin or vancomycin; metronidazole is no longer preferred.

Breast Cancer

MED-22 | Category: Oncology

Overview

Breast cancer is the most commonly diagnosed cancer in women worldwide. Risk factors include increasing age, female sex, family history, inherited mutations in BRCA1 and BRCA2 (with lifetime risks of about 55-72% and 45-69% respectively), early menarche, late menopause, nulliparity, hormone therapy, obesity after menopause, and alcohol use. Most invasive cancers are invasive ductal carcinoma, followed by invasive lobular carcinoma. Ductal carcinoma in situ (DCIS) is a non-invasive precursor.

Diagnosis and Evaluation

Screening mammography is recommended by the US Preventive Services Task Force every 2 years for women aged 40-74. Evaluation of a palpable mass includes clinical breast examination, imaging (ultrasound and diagnostic mammography), and core needle biopsy. Tumor assessment includes histologic grade, estrogen receptor (ER), progesterone receptor (PR), and HER2 status, which define molecular subtypes: luminal A and B (hormone-receptor positive), HER2-enriched, and triple-negative (ER, PR, and HER2 all negative), the latter being more aggressive. Staging uses the TNM system.

Management

Local treatment is breast-conserving surgery with radiation or mastectomy, with sentinel lymph node biopsy for axillary staging. Systemic therapy depends on subtype. Hormone-receptor-positive cancers receive endocrine therapy for 5-10 years: tamoxifen (selective estrogen receptor modulator, used pre- and postmenopausal) or aromatase inhibitors (anastrozole, letrozole, exemestane) in postmenopausal women; CDK4/6 inhibitors (palbociclib, ribociclib, abemaciclib) are added in advanced disease. HER2-positive cancers are treated with trastuzumab, pertuzumab, and trastuzumab emtansine or deruxtecan. Triple-negative disease is treated with chemotherapy and, in selected cases, pembrolizumab. Neoadjuvant therapy can downstage tumors.

Key Points

- Tamoxifen increases the risk of endometrial cancer and thromboembolism.
- Aromatase inhibitors can cause bone loss and arthralgias.
- Trastuzumab can cause cardiotoxicity; monitor LVEF.
- BRCA carriers may consider risk-reducing mastectomy and salpingo-oophorectomy.

Colorectal Cancer

MED-23 | Category: Oncology

Overview

Colorectal cancer (CRC) is a leading cause of cancer death. Most cases arise from adenomatous polyps through the adenoma-carcinoma sequence involving APC, KRAS, and TP53 mutations. Risk factors include age, family history, inflammatory bowel disease, obesity, diets high in processed or red meat, smoking, and alcohol. Hereditary syndromes include Lynch syndrome (mismatch repair gene mutations; hereditary nonpolyposis colorectal cancer) and familial adenomatous polyposis (APC mutation, near-100% cancer risk without colectomy).

Diagnosis and Evaluation

The USPSTF recommends average-risk screening from age 45 to 75 using colonoscopy every 10 years, annual fecal immunochemical test (FIT), stool DNA-FIT testing every 1-3 years, or CT colonography every 5 years. Symptoms include change in bowel habit, rectal bleeding, iron deficiency anemia, and abdominal pain; right-sided tumors often present with occult bleeding, and left-sided tumors with obstruction. Diagnosis is by colonoscopy with biopsy; staging uses CT of chest, abdomen, and pelvis, and the tumor marker carcinoembryonic antigen (CEA) is used to monitor recurrence rather than for screening.

Management

Surgery with adequate lymph node resection is the cornerstone of curative therapy. Adjuvant chemotherapy, typically FOLFOX or CAPOX (fluoropyrimidine plus oxaliplatin), is recommended for stage III colon cancer and selected high-risk stage II disease. Rectal cancer treatment often uses neoadjuvant chemoradiation or short-course radiation with total mesorectal excision. Metastatic disease is treated with combination chemotherapy plus targeted therapy based on molecular testing: anti-EGFR antibodies (cetuximab, panitumumab) only in RAS wild-type tumors, bevacizumab for angiogenesis, BRAF V600E-directed therapy, and immune checkpoint inhibitors (pembrolizumab) for microsatellite instability-high or mismatch repair-deficient tumors. Isolated liver metastases may be resectable.

Key Points

- Microsatellite instability testing is recommended for all new CRC diagnoses.
- Peripheral neuropathy is a dose-limiting toxicity of oxaliplatin.
- Aspirin may reduce CRC risk but is not a routine recommendation for screening substitution.
- 5-fluorouracil toxicity may be severe in DPYD-deficient patients.

Lung Cancer

MED-24 | Category: Oncology

Overview

Lung cancer is the leading cause of cancer death worldwide, with tobacco smoking responsible for about 80-85% of cases. Other risk factors include radon, asbestos, occupational carcinogens, air pollution, and prior chest radiation. It is divided into small cell lung cancer (SCLC, about 15%) and non-small cell lung cancer (NSCLC, about 85%), which includes adenocarcinoma (most common, peripheral, frequent in never-smokers), squamous cell carcinoma (central, associated with hypercalcemia from PTHrP), and large cell carcinoma.

Diagnosis and Evaluation

Annual low-dose CT screening is recommended for adults aged 50-80 with a 20 pack-year smoking history who currently smoke or quit within the past 15 years. Symptoms include persistent cough, hemoptysis, dyspnea, weight loss, and chest pain; paraneoplastic syndromes include SIADH and Cushing syndrome (SCLC), Lambert-Eaton myasthenic syndrome, and hypercalcemia (squamous). Pancoast tumors cause Horner syndrome and shoulder pain; superior vena cava syndrome causes facial swelling. Diagnosis requires tissue biopsy with molecular testing for EGFR, ALK, ROS1, BRAF, KRAS G12C, MET, RET, NTRK, and PD-L1 expression in NSCLC.

Management

Early-stage NSCLC is treated surgically (lobectomy) or with stereotactic body radiotherapy for non-operative patients, often with adjuvant therapy. Targeted therapy is chosen by mutation: EGFR-mutant disease responds to osimertinib; ALK-rearranged disease to alectinib or lorlatinib; KRAS G12C to sotorasib or adagrasib. Without actionable drivers, advanced NSCLC is treated with immune checkpoint inhibitors (pembrolizumab, atezolizumab, nivolumab) with or without platinum-based chemotherapy. Limited-stage SCLC is treated with concurrent chemoradiation (cisplatin or carboplatin plus etoposide) and prophylactic cranial irradiation in selected cases; extensive-stage SCLC uses chemotherapy plus immunotherapy (atezolizumab or durvalumab).

Key Points

- Smoking cessation improves outcomes at any stage.
- Immune checkpoint inhibitors can cause pneumonitis, colitis, hepatitis, and endocrinopathies.
- Small cell lung cancer is highly chemosensitive but relapses early.
- Brain metastases are common and treated with radiosurgery or targeted agents with CNS penetration.

Major Depressive Disorder

MED-25 | Category: Psychiatry

Overview

Major depressive disorder (MDD) is a common mental disorder characterized by persistent low mood or loss of interest (anhedonia) together with other symptoms for at least 2 weeks. DSM-5 requires five or more of the following symptoms, including at least one of depressed mood or anhedonia: depressed mood, diminished interest, weight or appetite change, insomnia or hypersomnia, psychomotor agitation or retardation, fatigue, worthlessness or guilt, impaired concentration, and recurrent thoughts of death or suicidal ideation. Symptoms must cause significant impairment and not be explained by substances, medical conditions such as hypothyroidism, or bipolar disorder.

Diagnosis and Evaluation

The Patient Health Questionnaire-9 (PHQ-9) screens and monitors severity (scores of 5, 10, 15, and 20 indicate mild, moderate, moderately severe, and severe depression). Evaluation includes suicide risk assessment, screening for bipolar disorder (history of mania or hypomania) before starting an antidepressant, substance use, and laboratory tests to exclude medical contributors such as thyroid dysfunction, anemia, and vitamin B12 deficiency.

Management

Mild-to-moderate depression is treated with psychotherapy (cognitive behavioral therapy, interpersonal therapy) and/or antidepressants; moderate-to-severe depression typically warrants combined treatment. First-line medications are selective serotonin reuptake inhibitors (sertraline, escitalopram, fluoxetine); alternatives include SNRIs (venlafaxine, duloxetine), bupropion, and mirtazapine. Response is usually seen after 2-6 weeks, and treatment continues for 6-12 months after remission. Risks include serotonin syndrome (agitation, hyperthermia, clonus, diarrhea) with combinations of serotonergic drugs and a boxed warning for increased suicidal thoughts in patients under 25. Abrupt discontinuation causes withdrawal symptoms. Treatment-resistant depression may respond to augmentation, esketamine nasal spray, transcranial magnetic stimulation, or electroconvulsive therapy, which is highly effective in severe, psychotic, or catatonic depression.

Key Points

- Lithium and bupropion carry specific risks: lithium toxicity (narrow therapeutic index) and bupropion lowers seizure threshold.
- MAO inhibitors require dietary tyramine restriction.
- SSRIs can cause sexual dysfunction and hyponatremia (SIADH) in older adults.
- Sudden emergency evaluation is needed for any patient with an active suicide plan.

Epilepsy

MED-26 | Category: Neurology

Overview

Epilepsy is a neurological disorder defined by a tendency to recurrent unprovoked seizures; it is diagnosed after two unprovoked seizures more than 24 hours apart, or after one seizure with a high (60% or greater) risk of recurrence. Seizures are classified as focal onset (with preserved or impaired awareness), generalized onset (tonic-clonic, absence, myoclonic, atonic), or unknown onset. Causes include structural lesions (stroke, tumor, trauma, cortical malformations), genetic syndromes, infection, autoimmune encephalitis, and metabolic disorders.

Diagnosis and Evaluation

Evaluation includes detailed history from witnesses, neurological examination, electroencephalography (EEG), and MRI of the brain. Typical absence seizures show 3-Hz spike-and-wave discharges on EEG and are triggered by hyperventilation. Status epilepticus is a seizure lasting 5 minutes or longer, or repeated seizures without recovery; it is a medical emergency. Differential diagnosis includes syncope, psychogenic nonepileptic events, migraine, and movement disorders.

Management

Antiseizure medication choice depends on seizure type and patient factors. Levetiracetam, lamotrigine, and carbamazepine/oxcarbazepine are commonly used for focal seizures; valproate, lamotrigine, and levetiracetam for generalized seizures, with ethosuximide first-line for childhood absence epilepsy. Valproate is highly teratogenic (neural tube defects and neurodevelopmental effects) and should be avoided in women of childbearing potential when possible. Carbamazepine requires HLA-B*1502 testing in patients of Asian ancestry due to risk of Stevens-Johnson syndrome; phenytoin causes gingival hyperplasia and hirsutism. Status epilepticus is treated with intravenous benzodiazepines (lorazepam) as first line, followed by a second-line agent (levetiracetam, fosphenytoin, or valproate). Refractory epilepsy may be treated with epilepsy surgery, vagus nerve stimulation, or ketogenic diet.

Key Points

- First-line emergency drug for seizure lasting over 5 minutes: benzodiazepine.
- Folic acid supplementation is advised for women with epilepsy who may become pregnant.
- Enzyme-inducing antiseizure drugs reduce effectiveness of oral contraceptives.
- Patients should follow local driving laws, which usually require a seizure-free interval.

Alzheimer Disease and Dementia

MED-27 | Category: Neurology

Overview

Dementia is an acquired decline in cognition severe enough to interfere with independence. Alzheimer disease (AD) is the most common cause (60-70%), followed by vascular dementia, dementia with Lewy bodies, and frontotemporal dementia. AD pathology comprises extracellular amyloid-beta plaques and intraneuronal neurofibrillary tangles of hyperphosphorylated tau, with early atrophy of the hippocampus and medial temporal lobe. Age is the strongest risk factor; the APOE e4 allele increases risk, while rare mutations in APP, PSEN1, and PSEN2 cause early-onset familial disease.

Diagnosis and Evaluation

Typical early symptoms include short-term memory loss, word-finding difficulty, disorientation, and impaired executive function, with later behavioral changes and loss of activities of daily living. Cognitive screening tools include the Mini-Mental State Examination (MMSE; maximum 30) and the Montreal Cognitive Assessment (MoCA; 26 or above is normal). Reversible causes should be excluded: vitamin B12 deficiency, hypothyroidism, depression, medication effects, normal-pressure hydrocephalus (the triad of gait disturbance, urinary incontinence, and dementia), and delirium. MRI shows atrophy; biomarkers such as plasma p-tau217, cerebrospinal fluid amyloid and tau, and amyloid PET can support the diagnosis.

Management

Symptomatic treatment includes cholinesterase inhibitors (donepezil, rivastigmine, galantamine) for mild-to-moderate disease and memantine, an NMDA receptor antagonist, for moderate-to-severe disease. Anti-amyloid monoclonal antibodies (lecanemab, donanemab) can modestly slow decline in early AD with confirmed amyloid pathology, with a risk of amyloid-related imaging abnormalities (ARIA), including edema and microhemorrhage, requiring MRI monitoring. Management also includes caregiver support, safety assessment, treatment of vascular risk factors, exercise, and nonpharmacologic approaches to agitation; antipsychotics carry a boxed warning for increased mortality in older adults with dementia. Dementia with Lewy bodies features visual hallucinations, parkinsonism, and fluctuating cognition, and patients are very sensitive to antipsychotics.

Key Points

- Cholinesterase inhibitor side effects: nausea, bradycardia, and syncope.
- Delirium is acute and fluctuating; dementia is chronic and progressive.
- Hearing loss, hypertension, and physical inactivity are modifiable dementia risk factors.
- Anticholinergic medications worsen cognition and should be avoided.

Rheumatoid Arthritis

MED-28 | Category: Rheumatology

Overview

Rheumatoid arthritis (RA) is a chronic systemic autoimmune disease causing symmetric inflammatory polyarthritis of small joints of the hands and feet, with synovial hyperplasia (pannus) that erodes cartilage and bone. It is more common in women, usually presents between ages 30 and 60, and is associated with HLA-DR4 (shared epitope), smoking, and PTPN22 variants. Extra-articular manifestations include rheumatoid nodules, interstitial lung disease, pleural effusion, scleritis, vasculitis, Felty syndrome (RA, splenomegaly, neutropenia), and increased cardiovascular risk.

Diagnosis and Evaluation

Patients report morning stiffness lasting more than 1 hour, swelling of metacarpophalangeal (MCP), proximal interphalangeal (PIP), and wrist joints, with sparing of distal interphalangeal joints. The 2010 ACR/EULAR classification uses joint involvement, serology (rheumatoid factor and anti-cyclic citrullinated peptide antibodies, anti-CCP being more specific), acute-phase reactants (ESR, CRP), and symptom duration of 6 weeks or more. Radiographs may show periarticular osteopenia, joint-space narrowing, and marginal erosions. Disease activity is tracked with DAS28 or CDAI.

Management

Early treatment with disease-modifying antirheumatic drugs (DMARDs) prevents joint damage ('treat to target' aiming for remission or low disease activity). Methotrexate is the anchor drug, given weekly with folic acid supplementation; monitor liver enzymes and blood counts and avoid in pregnancy, significant liver disease, and heavy alcohol use. Other conventional DMARDs include hydroxychloroquine, sulfasalazine, and leflunomide. If the response is inadequate, add a biologic (TNF inhibitors such as adalimumab, etanercept, infliximab; abatacept; tocilizumab, an anti-IL-6 receptor antibody; rituximab) or a JAK inhibitor (tofacitinib, baricitinib, upadacitinib). Screening for latent TB and hepatitis B is mandatory before biologics. Glucocorticoids provide bridging therapy at the lowest effective dose.

Key Points

- JAK inhibitors carry warnings for serious infections, thrombosis, malignancy, and major cardiovascular events.
- Rheumatoid factor can be positive in other diseases and healthy older adults.
- Atlantoaxial subluxation is a risk in long-standing RA before intubation.
- Hydroxychloroquine requires periodic retinal screening.

Osteoporosis

MED-29 | Category: Endocrine / Musculoskeletal

Overview

Osteoporosis is a skeletal disorder of reduced bone strength leading to increased fracture risk, mainly of the vertebrae, hip, and distal radius. It is most common in postmenopausal women because of estrogen loss, but also occurs in men and in secondary forms caused by glucocorticoids, hyperthyroidism, hyperparathyroidism, celiac disease, vitamin D deficiency, hypogonadism, and drugs such as aromatase inhibitors, proton pump inhibitors (long-term), and anticonvulsants. Risk factors include age, low body weight, smoking, alcohol, family history of hip fracture, and prior fragility fracture.

Diagnosis and Evaluation

Diagnosis is based on a prior fragility fracture of hip or vertebra, or bone mineral density (BMD) measured by dual-energy X-ray absorptiometry (DXA). WHO defines normal as a T-score of -1.0 or higher, osteopenia (low bone mass) as -1.0 to -2.5, and osteoporosis as -2.5 or lower. The FRAX tool estimates 10-year probability of major osteoporotic and hip fracture. Screening with DXA is recommended for women aged 65 and older, and for younger postmenopausal women at increased risk. Evaluation of secondary causes includes calcium, vitamin D, parathyroid hormone, thyroid function, and kidney function.

Management

Non-pharmacologic measures include weight-bearing exercise, fall prevention, adequate calcium (about 1,000-1,200 mg per day) and vitamin D (600-800 IU per day), smoking cessation, and moderate alcohol intake. First-line drug therapy is an oral bisphosphonate (alendronate, risedronate), taken fasting with a full glass of water while remaining upright for 30 minutes to avoid esophagitis; intravenous zoledronic acid is annual. Rare adverse effects include osteonecrosis of the jaw and atypical femoral fractures with prolonged use, so drug holidays are considered after 3-5 years in lower-risk patients. Denosumab (RANKL inhibitor) must be continued or followed by another agent to avoid rebound vertebral fractures. Anabolic agents (teriparatide, abaloparatide, romosozumab) are used for very high-risk patients.

Key Points

- Treat if T-score is -2.5 or lower, prior hip or vertebral fracture, or high FRAX risk with osteopenia.
- Glucocorticoid-induced osteoporosis risk begins with 3 months of prednisone at 5 mg/day or more.
- Vitamin D deficiency should be corrected before starting antiresorptives to avoid hypocalcemia.
- Romosozumab has a cardiovascular warning.

Thyroid Disorders: Hypothyroidism and Hyperthyroidism

MED-30 | Category: Endocrine

Overview

Hypothyroidism is thyroid hormone deficiency; the commonest cause in iodine-sufficient regions is Hashimoto thyroiditis (autoimmune, with anti-thyroid peroxidase antibodies). Hyperthyroidism is excess thyroid hormone; the commonest cause is Graves disease (TSH receptor-stimulating antibodies), followed by toxic multinodular goiter and toxic adenoma. Thyroiditis can cause transient thyrotoxicosis followed by hypothyroidism. Pregnancy and the postpartum period are times of altered thyroid function.

Diagnosis and Evaluation

In primary hypothyroidism TSH is elevated with low free T4; symptoms include fatigue, cold intolerance, weight gain, constipation, dry skin, bradycardia, and delayed relaxation of deep tendon reflexes. In primary hyperthyroidism TSH is suppressed with elevated free T4 and/or T3; symptoms include weight loss, heat intolerance, tremor, palpitations, anxiety, and diarrhea. Graves disease additionally causes diffuse goiter, ophthalmopathy (exophthalmos), and pretibial myxedema. Radioactive iodine uptake scan distinguishes diffuse high uptake (Graves) from low uptake (thyroiditis). Subclinical hypothyroidism is elevated TSH with normal free T4.

Management

Hypothyroidism is treated with oral levothyroxine (typically 1.6 micrograms/kg/day, taken on an empty stomach, separated from calcium and iron by 4 hours), with TSH measured 6-8 weeks after dose changes; lower doses are used in elderly patients or with coronary disease. Hyperthyroidism is treated with antithyroid drugs (methimazole; propylthiouracil preferred in the first trimester of pregnancy and thyroid storm), beta-blockers for symptoms, radioactive iodine ablation, or thyroidectomy. Methimazole may cause agranulocytosis (warn patients to report fever or sore throat) and propylthiouracil may cause severe liver injury. Thyroid storm is a life-threatening emergency treated with beta-blockers, antithyroid drugs, iodine solution (given after thioamide), and glucocorticoids. Myxedema coma, severe decompensated hypothyroidism, is treated with intravenous levothyroxine and hydrocortisone.

Key Points

- Radioactive iodine is contraindicated in pregnancy and breastfeeding.
- Radioactive iodine can worsen Graves ophthalmopathy; smoking increases this risk.
- Untreated maternal hypothyroidism increases miscarriage and neurocognitive risk in the offspring.
- Amiodarone can induce either hypo- or hyperthyroidism.